Andrew Strachan

M.S. Cybersecurity Candidate | CyberAICorps Scholar | UAB
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SFS CyberAICorps scholarship recipient (NSF-funded, with a post-graduation government service commitment) Willing to obtain a security clearance.

Summary

SFS CyberAICorps scholarship recipient (NSF-funded, with a post-graduation government service commitment), pursuing an M.S. in Cybersecurity at UAB (GPA 3.75). M.S. Cybersecurity student and builder. Combines systems programming and zero-trust engineering with executive compliance, industrial cGMP quality controls, and award-winning career and technical education leadership.

Education

University of Alabama at Birmingham January 2026 – Present (Expected Graduation: Dec 2027)
Master of Science in Cybersecurity
GPA / standing: 3.75 / 4.0
SFS CyberAICorps scholarship recipient (NSF-funded, with a post-graduation government service commitment)

University of Montevallo Fall 2024 – Spring 2025
Provisional Certificate in a Teaching Field: Business, Marketing, and Finance (non-degree)
GPA / standing: 3.66 / 4.0
ALSDE State CTE Certification · NAOF Curriculum Specialization

Mississippi College August 2011 – May 2016
Bachelor of Science (B.S.)
GPA / standing: 3.50 / 4.0
cum laude · Delta Epsilon Iota Academic Honor Society · Phi Mu Alpha Sinfonia

University of Mississippi School of Medicine January 2016 – July 2020
Medical Studies, Course and clinical work, University of Mississippi School of Medicine, 2016–2020
GPA / standing: Coursework Completed in Good Standing
Virginia Covington Award (2017) · President & Founder: P.A.L.S. (Peer-Assisted Learning Society) · Chair: UMMC/MBN Opioid Crisis Council · Human Anatomy Prosector (May–Sep 2017)

Skills

Cybersecurity & Security Architecture: STRIDE Threat Modeling, Zero-Trust Architecture, NIST SP 800-53 / CSF Compliance, Network Defense & Packet Analysis (Wireshark, Nmap), Cryptographic Hashing & SHA-256 Block Verification, Cloud Computing (Summer 2026), CS 532 Systems Programming, Identity & Access Management (IAM / Least Privilege), Firewall, Proxy & Access Control Lists (ACLs)

Artificial Intelligence & Local Models: On-Device Retrieval-Augmented Generation (Local RAG), Apple MLX & Metal Performance Shaders, Quantized LLMs (Ollama / Gemma 4 / Llama 3), SwiftData & Local SQLite Vector Indexing, Model Context Protocol (MCP) Sandboxed Tool Routing, Agentic AI Workflow Architecture & Permission Gates, Prompt Engineering & Automated Evaluation Pipelines, Knowledge Graphs for Enterprise GenAI

Cloud & DevSecOps: AWS (S3, CloudFront, Lambda, DynamoDB, SES), Firebase (Hosting, Firestore, Cloud Functions), Google Cloud Platform Core Services, Docker Container Isolation & Sandboxing, Expiring HMAC Token Verification & Secure File Transfer, CI/CD Pipeline Security & Static Analysis (SAST), Vercel & Next.js Server-Side Deployment

Systems & Programming Languages: Swift / SwiftUI (Native macOS & iOS), Python (Systems, Data Analytics, Cryptography), Java 21 (Zero-Trust Sockets & Agent Gatekeepers), TypeScript / JavaScript (ESNext, React 19, Vite), C & Assembly Systems Fundamentals, SQL (SQLite, PostgreSQL, Relational Schema Design), Godot GDScript & WebGL Spatial Computing

Leadership & Educational Instruction: Microsoft Certified Educator (MCE Credly-Verified), Career and Technical Education (CTE) Curriculum Design, Student Organization Advising & Competitive Coaching, DECA State-Level Competition Preparation, TAPE Standards Compliance & Educational Governance, Physical Computing & STEM Robotics (BBC micro:bit), Cross-Disciplinary Team Facilitation & Executive Presentation

Experience

Python Coding Summer Camp Teaching Assistant, UAB Jun 2026 – Jul 2026
University of Alabama at Birmingham (Department of Computer Science) · Birmingham, AL

- Instructed middle school cohorts in Python programming fundamentals, algorithmic loops, and physical computing.
- Guided students to interface BBC micro:bit microcontrollers with LED displays, motion sensors, and external robotics.
- Mentored students in scripting custom autonomous flight trajectories for aerial drones using Python.

Career & Technical Education (CTE) Teacher — Business, Marketing & Finance July 2024 – June 2025 (Full-Time, 40 hrs/wk)
Shades Valley High School (Jefferson County Board of Education - JEFCOED) · Birmingham, AL

- Taught Business Finance, Customer Service, and Sales in the premier National Academy of Finance (NAOF) program.
- Served as official Chapter Adviser for the state's largest student career and technical business organization.
- Head Coach for the Shades Valley High School Esports Team; raised funding and secured corporate sponsorships.

Career & Technical Education (CTE) Teacher — Business, Marketing & Finance (DECA Chapter Founder & State Competition Coach) August 2023 – June 2024 (Full-Time, 40 hrs/wk)
Corner High School (Jefferson County Board of Education - JEFCOED) · Dora, AL

- Founded, chartered, and advised a brand-new DECA chapter, opening statewide competitive business leadership to rural students.
- Coached student competitors in business finance and marketing, achieving 1st place and 2nd place state trophies at the 2024 Alabama DECA Career Development Conference.
- Engineered experiential business curricula in financial modeling, spreadsheet automation, and entrepreneurship.

Operations Director January 2023 – May 2023
MidSouth Extracts LLC · Tupelo, MS

- MidSouth Extracts, a Mississippi Medical Cannabis Program (MSDH)-licensed processing facility, needed its MSDH processing license, FDA food-facility registration, and required state and county licenses.
- Personally procured its MSDH (Mississippi Medical Cannabis Program) processing license, FDA food-facility registration, and all other required state and county licenses, including its MSDH manufactured-food permit and City of Tupelo local permits; wrote its 11-SOP library and HACCP program.

Founder & Principal Technologist January 2022 – Present (15–20 hrs/wk)
MaQkrs Consulting · Birmingham, AL & Clinton, MS

- Delivered technical advisory services on zero-trust digital infrastructure, FDA/cGMP compliance, and secure cloud deployments.
- Developed custom client platforms including e-commerce storefronts, clinical healthcare portals, and local AI workflows.
- Mentored student interns in security baseline auditing, vulnerability scanning, and cloud least-privilege configurations.

Insurance Sales Representative, SelectQuote Insurance Services, Clinton, MS, Apr 2021 – Sep 2021 Apr 2021 – Sep 2021
SelectQuote Insurance Services · Clinton, MS

- Awarded the SelectQuote Top Sales Award (2021).

Team Lead, Oct 2021 – Mar 2022 Oct 2021 – Mar 2022
SelectQuote Insurance Services · Clinton, MS

- Awarded the SQ Team Lead Award (2021).

Sales Development Specialist (built and led the new-rep training program), Apr 2022 – Nov 2022 Apr 2022 – Nov 2022
SelectQuote Insurance Services · Clinton, MS

- Built and led the new-rep training program.
- Trained, onboarded, and coached incoming sales cohorts across life and health insurance divisions.
- Engineered micro-training modules and compliance scripts that reduced regulatory complaints and elevated retention.

Certified Village Health Team (VHT) member, UMMC & OmniMed, Uganda (Jul 2017 – Aug 2017) Jul 2017 – Aug 2017
UMMC & OmniMed · Uganda

- Mobilized and ran mobile clinics with the UMMC team; helped establish community health teaching groups.

Youth Leadership & Civic Mentorship August 2011 – Present (10 hrs/wk)
First Presbyterian Church · Gulfport, MS & Clinton, MS

- Provided sustained youth leadership, academic mentoring, and civic guidance across 15+ years of continuous service.
- Organized community outreach initiatives, disaster relief volunteering, and student developmental programs.
- Facilitated small-group discussions and character development for youth facing academic and personal challenges.

Human Anatomy Prosector (May–Sep 2017) May 2017 – September 2017
University of Mississippi School of Medicine (UMMC) · Jackson, MS

- Performed intricate anatomical dissections of human cadaveric donors to create master teaching specimens for medical and dental schools.
- Delivered lecture demonstrations to 100+ first-year medical students, teaching complex neurovascular and musculoskeletal relationships.

Human Gross Anatomy Laboratory Teaching Assistant August 2015 – May 2016
Mississippi College (Department of Biological Sciences) · Clinton, MS

- Facilitated laboratory dissections and practical exam tutorials for undergraduate pre-medical cohorts across two semesters.
- Enforced biosafety standards and laboratory chain of custody for chemical preservation protocols.